

RoboSax Melody Slot Machine

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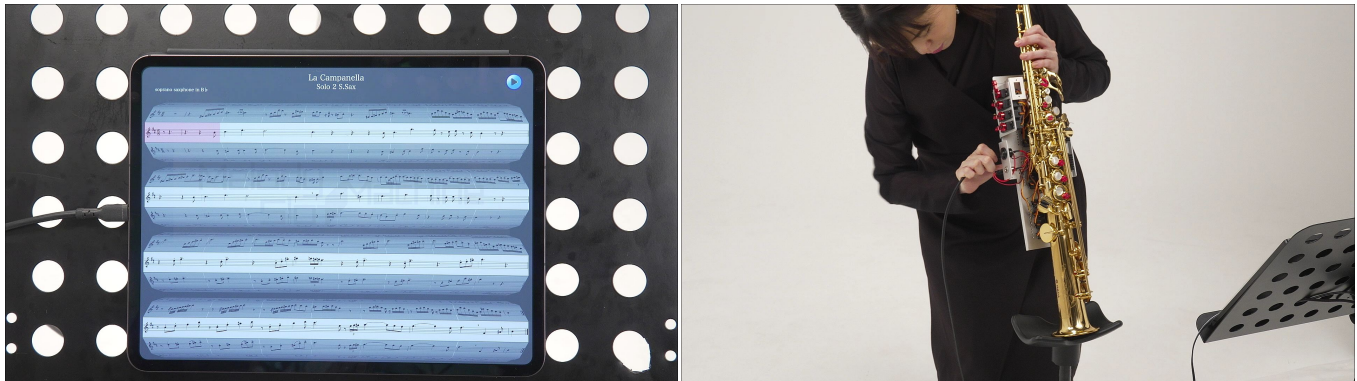


Figure 1: Melody Slot Machine and RoboSax

Abstract

This paper describes RoboSax Melody Slot Machine, which automatically plays fingering melodies selected using a dial-like controller with stave notation on a tablet screen. All that are required for the saxophone to output the melody selected by the tablet operator are the blowing and tonguing of the saxophonist. RoboSax Melody Slot Machine makes the task of selecting melodies, which was previously possible only for composers, possible for the general people and enables saxophone playing by those who cannot quickly read the note names on the staff.

CCS Concepts

• Applied computing → Sound and music computing.

Keywords

Generative Theory of Tonal Music (GTTM), Musical structure, Melody morphing method, Automatic fingering, Servomotor

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1 Introduction

Our goal is to make musical activities accessible to even the musical novice. We considered composition and performance as such musical activities. Many music novices have only passive experience of listening to music. In contrast, RoboSax Melody Slot Machine, which combines Melody Slot Machine and RoboSax, provides an active experience of controlling the music and performance in real time (Fig. 1). Melody Slot Machine can create approximately 7.9 septillion ($\approx 6^{32}$) combinations of melodies, with each of the 32 bars having 6 variations, and there are numerous first-time combinations of melodies that have never been played before. The users can then listen to a live saxophone performance of the melody of the combination they selected.

2 Melody Slot Machine

A composer can efficiently create a consistent melody on the basis of a variety of musical knowledge, but a musical novice cannot do so, because they do not have musical knowledge. However, when there are candidates for a melody, it is possible to choose one that one prefers. For example, Mozart composed a piece called "musical dice game" [7]. Mozart created 11 one-measure-long melodies for each of the 16 bars, and the piece was completed by selecting bars according to the sum of dice rolls. Even for a composer, it is difficult to create variations in melody so that consistency is maintained no matter which measure is selected.

Consistent melody variations are generated with Melody Slot Machine [1, 6] on the basis of the Generative Theory of Tonal Music (GTTM) [8]. The GTTM enables analysis of grouping structures, beat structures, and time-span trees from melodies. We were the first to implement GTTM analysis on a computer and introduced deep learning into the analysis, which dramatically improved of the analysis [2, 5]. The melodies of the variations are generated using a melody-morphing method on the basis of the GTTM (Fig. 2)

[3, 4]. Using this method, a melody that is intermediate between two melodies can be obtained from the two melodies through arithmetic operations.

Melody Slot Machine introduces a dial-type interface for enjoying the selection of variations. All that is required to experience composing with Melody Slot Machine is to rotate the dial.

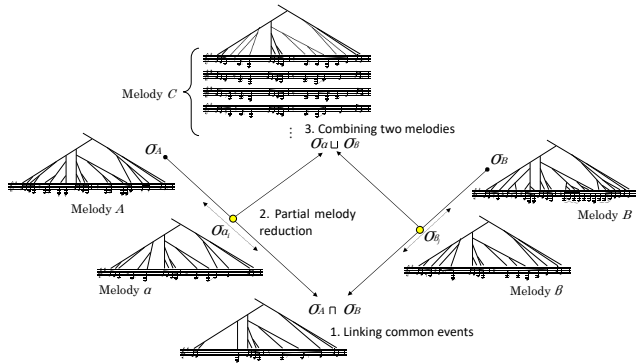


Figure 2: Melody-morphing method

3 RoboSax

Wind players finger, blow, and tongue to the rhythm of the piece while anticipating the score. Even experienced wind instrument players find it difficult to perform on a first-time score, but RoboSax makes it possible for even music novices to experience playing the saxophone by automating the process from reading the score to fingering. There are pianos on the market that play automatically, but they are all automated [9]. In contrast, RoboSax aims to reduce the burden on the performer by automating some of the movements necessary for performance. The microcontroller in RoboSax receives Musical Instrument Digital Interface (MIDI) signals output from Melody Slot Machine and drives servomotors (Fig. 3). The servomotor arm is connected to the saxophone key by a wire. When the motor rotates, the wire is pulled and the key moves to block or open the tone hole. All that are needed for the performer to experience playing a melody with RoboSax are blowing and tonguing.

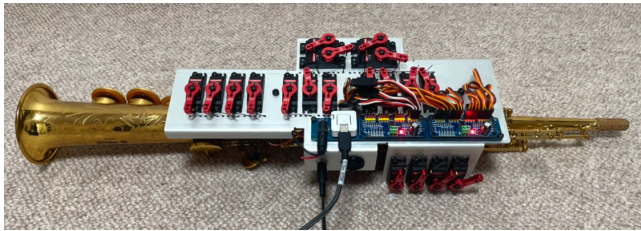


Figure 3: Servomotors

4 Experimental Results

To enable performance with the RoboSax Melody Slot Machine, the angle of the RoboSax servomotors were adjusted, and the timing of

sending MIDI signals from Melody Slot Machine to RoboSax was adjusted.

4.1 Servomotor Angle Adjustment

RoboSax is equipped with 19 FH-D9257 servomotors from Hobbyant Co. We adjusted the angle of each of the 19 servomotors. We have implemented a program for adjusting the angle of rotation when the tone holes are closed and open. If the difference in angles is too large, the current may momentarily exceed 500 mA and cause failure. Therefore, we adjusted the holes to open and close tightly and made sure that the difference in angles was not too large. After adjusting the 19 servomotors, the maximum difference between the open and closed angles was 34.3 degrees, minimum difference was 18.1 degrees, and average was 23.9 degrees.

4.2 Timing Adjustment

Melody Slot Machine is implemented on the iPad, and when connected to the RoboSax microcontroller via a USB 2.0 (Type-C) cable, the iPad recognizes RoboSax as a MIDI device and MIDI signals can be sent from Melody Slot Machine to RoboSax. Melody Slot Machine simultaneously outputs an acoustic signal for accompaniment and MIDI signal to drive RoboSax. A performer of RoboSax then plays along with the accompaniment track output from Melody Slot Machine. It takes some time for the MIDI signal to be sent to RoboSax and for the servomotors to move to open or close the tone holes. Therefore, if the iPad sends the MIDI signal at the same timing as the acoustic signal, the sound from the RoboSax performance will be slightly delayed.

We therefore implemented a function to send MIDI signals earlier than acoustic signals for a specified time to Melody Slot Machine so that the timing of acoustic signals and saxophone performances will match. When one professional saxophonist played a piece that she had practiced on a regular soprano saxophone for three days and varied the time to match her own timing while playing it on RoboSax, she felt that her timing was correct when the MIDI signal was sent 30 milliseconds ahead of the acoustic signal. The tempo of the song used for timing adjustment was 120 beats per minute, and the shortest note was a 32nd note.

5 Conclusion

This paper describes the RoboSax Melody Slot Machine, which allows even music novices to control combination of melodies and performances. The melody variations used in Melody Slot Machine use a structure based on GTTM, so that even if the variations are changed, the overall structure is maintained and the pieces does not break up. RoboSax controls 19 servomotors with a microcomputer based on MIDI signals received from Melody Slot Machine to execute fingerings automatically. By adjusting the angle of rotation of the RoboSax servomotors, the tone holes could be opened and closed reliably. By accelerating the MIDI signal sent from Melody Slot Machine to the RoboSax 30 milliseconds earlier than the acoustic signal for accompaniment, the professional saxophonist was able to play with uncomfortable timing. We will increase opportunities to perform with the RoboSax Melody Slot machine in the future and make improvements if any problems are found.

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